Quiz : 12

Q9:- $$\int \frac{x(3-5x)}{(3x+1)(x-1)^2}\,dx$$

$$= \int \frac{x(3-5x)}{(3x-1)(x-1)^2}\,dx$$

$$\frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(3x-1)} = \frac{x(3-5x)}{(3x-1)(x-1)^2}$$

$$A(x-1)(3x+1) + B(3x-1) + C(x-1)^2 = x(3-5x)$$

$$A(3x^2-3x+x-1) + 3Bx - B + C(x^2-2x+1) = 3x-5x^2$$

$$\Rightarrow \left.\begin{array}{l} 3A+C=-5 \\ -2A+3B-2C=3 \\ -A-B+C=0 \end{array}\right\} \rightarrow \begin{array}{l} -2A+3B-2C=3 \\ -3A-3B+3C=0 \\ \hline -5A+C=3 \end{array}$$

$$\begin{array}{l} 3A+C=-5 \\ -5A+C=3 \\ \hline 8A=-8 \end{array} \Rightarrow \boxed{A=-1}$$

$$3A+C=-5$$
$$-3+C=-5$$
$$\boxed{C=-2}$$

$$\Rightarrow -A-B+C=0 \Rightarrow 1-B-2=0 \Rightarrow \boxed{B=-1}$$

$$= \int \left[\frac{-1}{(x-1)} - \frac{1}{(x-1)^2} - \frac{2}{(3x-1)} \right] dx$$

$(3x-1)$
$\int u^{-1} \frac{du}{3}$
$du = 3\,dx$

$$= -\ln|x-1| + \frac{1}{(x-1)} - 2\int \frac{1}{(3x-1)}\,dx$$

$$= -\ln|x+1| + \frac{1}{(x-1)} - 2\int \frac{1}{u}\left(\frac{du}{3}\right)$$

$$= -\ln|x-1| + \frac{1}{(x-1)} - \frac{2}{3}\ln|3x-1| + C$$

C10:

(A) $\int \frac{\sin(1/x)}{x^2}\,dx$

Let $u = 1/x$, $du = \frac{1}{x^2}\,dx$

$$\int \sin(u)\,du = -\cos(u) + C$$

$$= -\cos(1/x) + C$$

(B) $\int x^2 \ln(x+1)\,dx$

Let $u = \ln(x+1)$, $du = 1/(x+1)\,dx$ | $dv = x^2$, $v = x^3/3$

$$= \frac{x^3\ln(x+1)}{3} - \frac{1}{3}\int \frac{x^3}{(x+1)}\,dx$$

=

(c) $\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx$

$= \int \sec^2 x \, dx - \int dx$

$= \tan x - x + C$

(C11) $$\int_2^{\infty} \frac{\ln(x)}{x^2}\,dx$$

$$= \lim_{b\to\infty} \int_2^{b} \frac{\ln(x)}{x^2}\,dx$$

Let $u = \ln(x)$, $dv = \frac{1}{x^2}\,dx$

$du = \frac{1}{x}\,dx$, $v = -\frac{1}{x}$

$$= \lim_{b\to\infty} \left[\frac{-\ln(x)}{x} - \int \left(-\frac{1}{x}\right)\left(\frac{1}{x}\right) dx\right]_2^{b}$$

$$= \lim_{b\to\infty} \left[-\frac{\ln(x)}{x} + \int \frac{1}{x^2}\,dx\right]_2^{b}$$

$$= \lim_{b\to\infty} \left[\frac{-\ln(x)}{x} - \left(\frac{1}{x}\right)\right]_2^{b}$$

$$= \lim_{b\to\infty} \left[\frac{-\ln(x) - 1}{x}\right]_2^{b}$$

$$= \lim_{b\to\infty} \left[\frac{-\ln(b) - 1}{b}\right] - \left[\frac{-\ln(2) - 1}{2}\right]$$

$$= \infty$$

$$\int_2^{\infty} \frac{\ln(x)}{x^2}\,dx = \infty$$

∴ The integral is diverges.